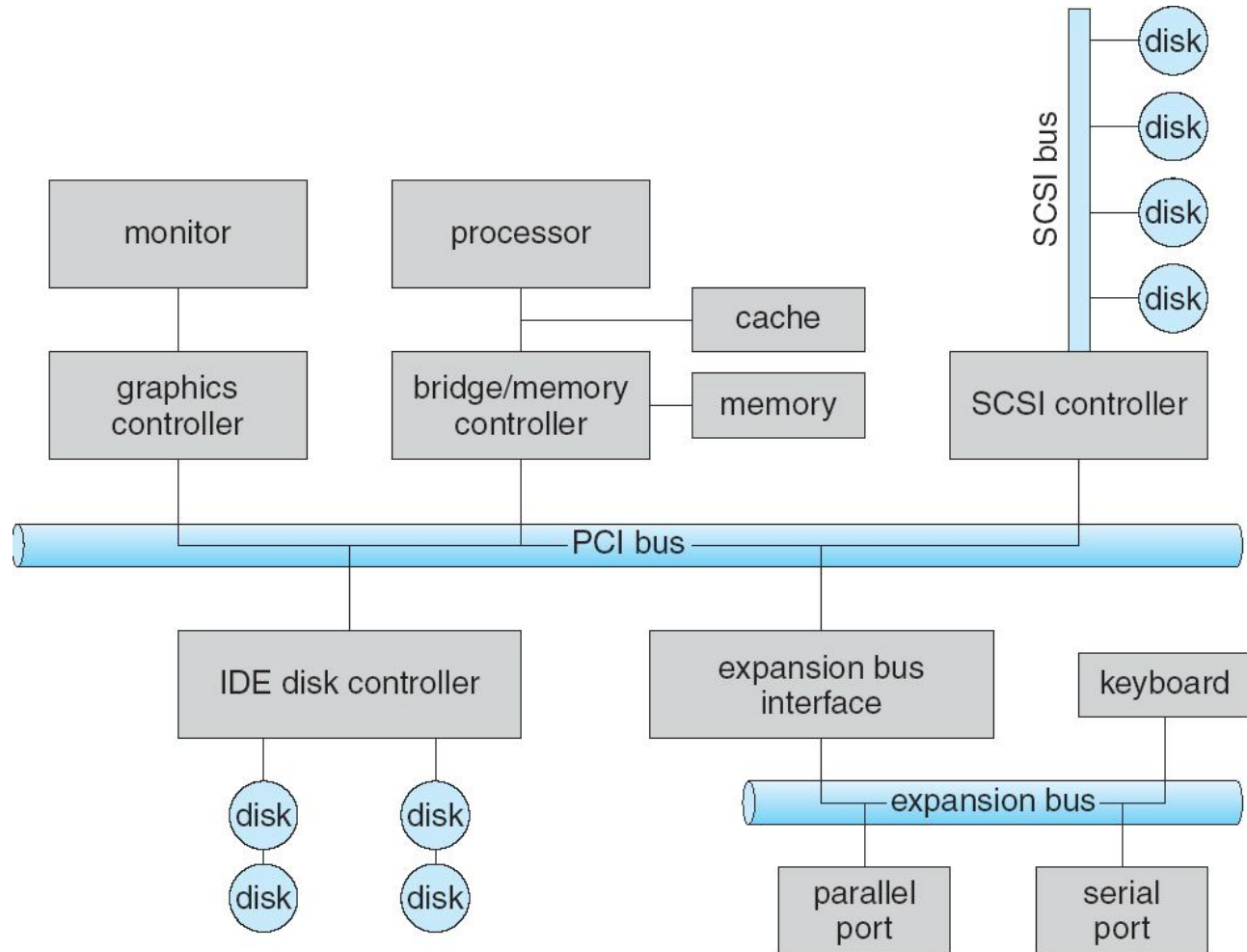


CSc 360
Operating Systems
I/O Systems

Wenjun Yang
Summer 2026

I/O structure

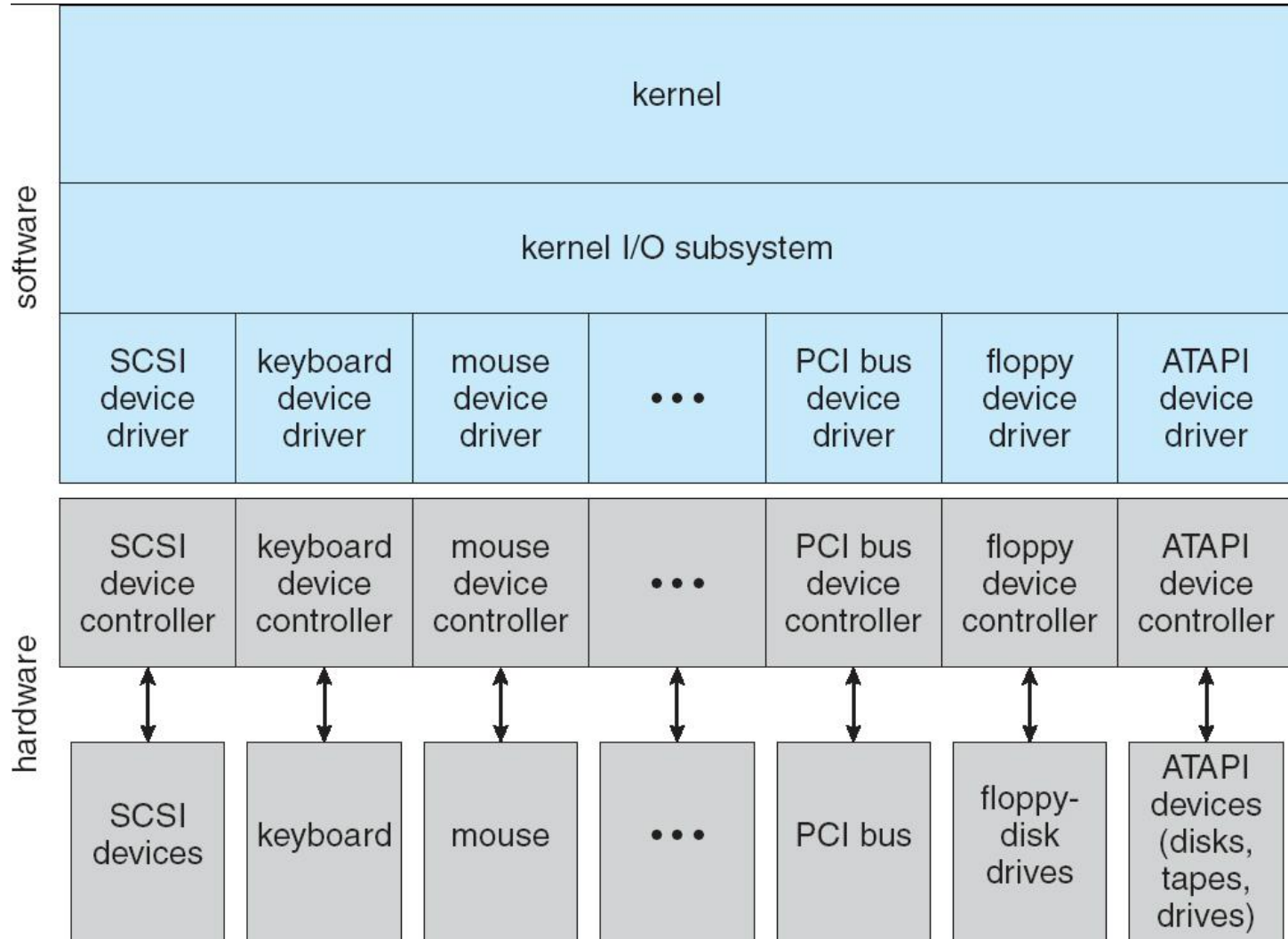
- I/O access
 - port number
 - interrupt #
 - DMA number
- E.g.,
 - PC



I/O access

- Polling: busy, ready, error
 - “busy-wait”
- Interrupt: event-driven
 - asynchronous operation
 - interrupt controller
- DMA: bulk data transfer
 - high-speed I/O
 - DMA controller

Kernel I/O structure



I/O characteristics

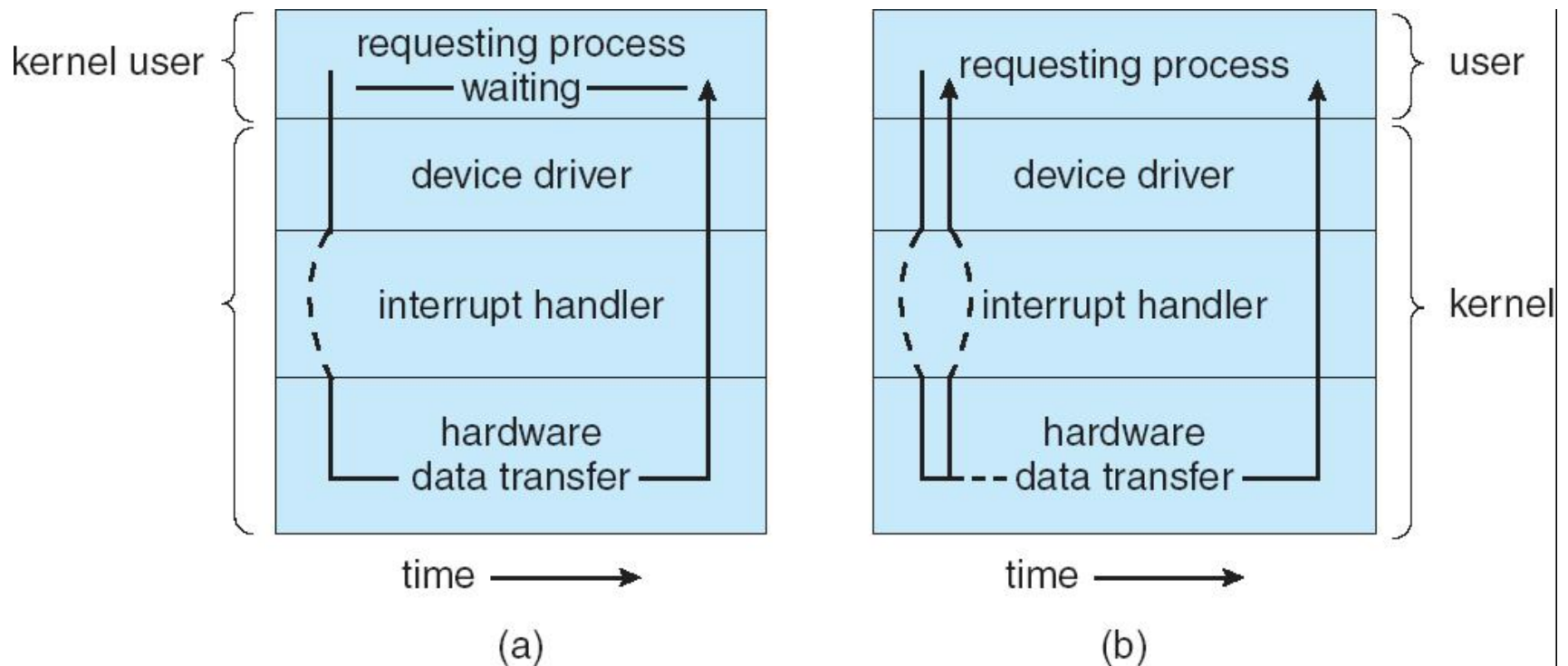
aspect	variation	example
data-transfer mode	character block	terminal disk
access method	sequential random	modem CD-ROM
transfer schedule	synchronous asynchronous	tape keyboard
sharing	dedicated sharable	tape keyboard
device speed	latency seek time transfer rate delay between operations	
I/O direction	read only write only read–write	CD-ROM graphics controller disk

Common devices

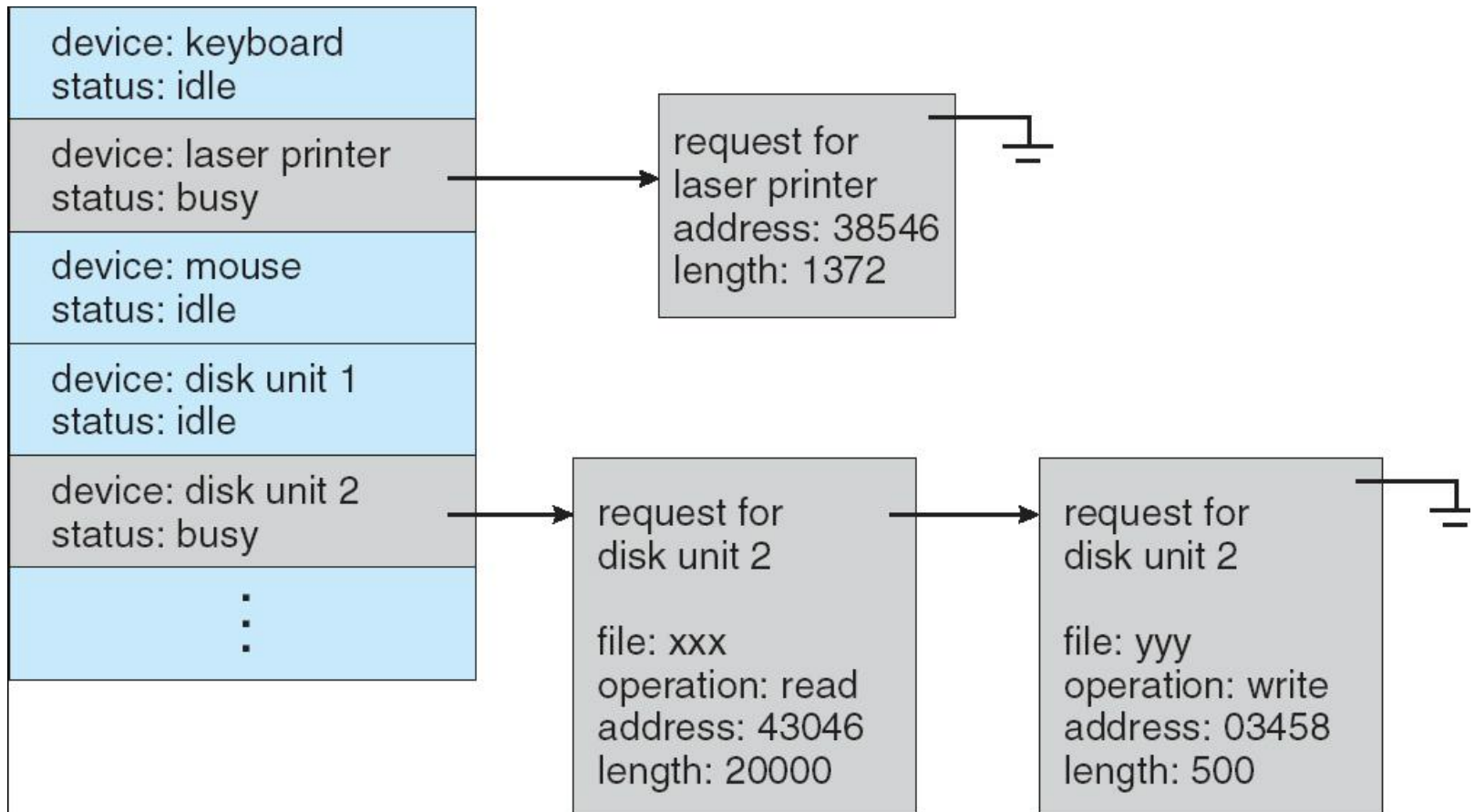
- Block device
 - e.g., disk drive
 - random access: read, write, seek
- Character device
 - e.g., keyboard, serial port
 - sequential access: `getc`, `putc`
- Network device
 - e.g., socket

Asynchronous I/O

- Also, blocking vs non-blocking I/O



I/O scheduling



More on I/O systems

- Caching
 - I/O is relatively slow
- Spooling
 - one job at a time
- Reservation
 - be aware of deadlock
- Error handling
- Protection: I/O access is privileged

This lecture

- I/O systems
 - I/O structures
 - I/O devices
 - block, character, network, etc
 - I/O access
 - polling, interrupt, DMA
- Again, the main focus of this course
 - process, memory, storage management!

Next lecture

- Term review
 - for the whole course
 - also in preparation for M3 (Dec 1)
 - course experience survey (CES)
<http://ces.uvic.ca>